Notes on Delirium

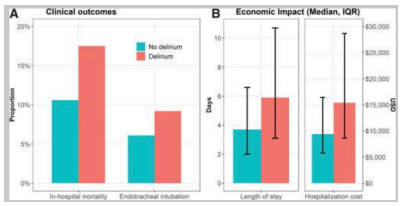
▼ Some extra thoughts

- I quite like the idea of aiming the pitch towards less developed countries, as we know delirium is much less detected in these areas
- It's also a lot more attainable to achieve a MVP in a weekend if we tailor to these regions
- So I need to find some papers looking at the cost/prevalence of delirium in these areas
- I think a nice point to think of in terms of why we need to focus on non-pharm prevention methods is, whilst yes antipsychotics can help with managing delirium, they have severe side effects like an increased likelihood of CV events

- one in three adults and children experience delirium during ICU stay
- Prevalence among residents living in long term care facilities is as high as 70%
- Delirium related adverse outcomes → decreased mobility and independence in activities of daily living, inc risk of mortality
- Causes distress among pt and their relatives/caregivers, also complicates the work of HCP
- Delirium is mostly preventable and is clinically diagnosed
- The Hospital Elder Life Programme (HELP) and the modified HELP programme has been designed to prevent and manage delirium → show lower odds of developing delirium and falls
- Systematic review all over the world show that nurses under-recognise delirium
- E.g. in Canada a study showed that in seven academic hospitals, medical and nursing staff only recognised delirium in half of the older adults with delirium in the ED
- Similar findings found in long-term care settings, attributed to lack of knowledge and detection skills
- In these residential care facilities it is easily overlooked as delirium is often superimposed on dementia or depression
- As it's a fluctuating mental state, assessments need to be performed frequently
- 4AT accuracy- sensitivity was 0.88 and specificity was 0.88 in a meta analysis

Study on medical and economic burden of delirium on hospitalisation:

- Study looked at effect-magnitude of delirium on acute respiratory failure
- Split pt up to delirium vs no delirium groups for ARF hospitalisations
- There was a racial disparity among African Americans
- Delirious pt had more co-morbidities, higher mortality and intubation rates
- Delirious pt have a longer length of stay and higher hospitalisation costs (5.9 days and $15,395 USD vs 3.7 days and $9393 USD in "No delirium" [$P < .001$], respectively)
- Patients in small non-teaching hospitals were diagnosed with delirium less freq compared to large urban, teaching centres
- Delirium is a significant burden on the healthcare system, about 82.4 billion dollars annually in the US
- Delirium is a reversible condition, TF hospital outcomes of primary medical illnesses are anticipated to improve if delirium is prevented and/or treated early
- study showed that delirium is associated with a 90% increase in overall hospital mortality, 70% increase in intubation rate, 59.5% increase in LOS, and 64% increase in hospitalization costs



2023 study on delirium in the US:

- Inc future risk of dementia
- It compromises pt safety by inc risk of falls, aspiration events and use of physical and chemical restraints
- Delirium is estimated to cost over 164 billion dollars in annually in US healthcare expenditures
- It's often missed due to inconsistent adoption of best practices to prevent and manage delirium and limited use of validated delirium screening tools.
- As part of World Delirium Awareness Day- March 15th 2023, a one day prevalence study was carried out in the US, it's the largest study to report on the clinically documented prevalence of delirium in the US
- 91 hospital units from all regions of the country participated
- In 72.2% of the 91 units, delirium was assessed twice daily
- 97.8% of the time nurses were primarily responsible for assessing delirium
- In non ICU the Confusion Assessment Method and the Nursing Delirium Screening Scale were most freq used as delirium screening tools
- Top 3 non pharmacologic interventions across units were pain management (95.6%), verbal re-orientation (94.5%), and mobilization (87.9%),
- top three pharmacological interventions were: reduction of deliriogenic drugs (46.2%), followed by melatonin use (39.5%), and evaluation of drugs by a specialist (38.5%, i.e., consultation by a delirium expert, which could include a consultant-liaison psychiatrist, neurologist, geriatrician, pharmacist, nursing, etc.).
- The top three reported delirium-related education and processes across units were educational training about delirium (65.9%), delirium mentioned in handovers (52.8%), and communication of delirium screening rate on the unit (49.5%; Table 2). The non-ICU setting reported a significantly higher use of delirium flyers for staff than did ICU units (non-ICU: 60.6% vs ICU: 31%, $P = 0.006$) and a significantly higher presence of dedicated delirium experts known by staff (non-ICU: 42.4% vs ICU: 19%, $P = 0.016$).
- The top barriers to the implementation of delirium care and/or use of evidence-based strategies across units were the shortage of personnel/staff (64.8%), lack of awareness (56.0%), and lack of time to educate and train staff (51.7%).
- Significant differences between non-ICU and ICU settings included the following: non-ICU settings more frequently reported lack of awareness (69.7% vs 48.3%, $P = 0.048$), missing knowledge about delirium (60.6% vs 36.2%, $P = 0.025$), and lack of appropriate scores for assessment of delirium (24.2% vs 8.6%, $P = 0.041$) compared to the ICU. In contrast, ICU settings reported more frequently than non-ICU settings that patients were difficult to assess (ICU 50% vs non-ICU 27.3%, $P = 0.035$), and other problems were more challenging (ICU 29.3% vs non-ICU 9.1%, $P = 0.034$)

	Overall n=91	Non-ICU N=33	ICU N=58	*P* value
Shortage of personnel/staff	59 (64.8)	24 (72.7)	35 (60.3)	0.234
Lack of awareness	51 (56.0)	23 (69.7)	28 (48.3)	**0.048**
Lack of time to educate and train staff	47 (51.7)	17 (51.5)	30 (51.7)	0.985
Communication gaps between professions	46 (50.6)	15 (45.5)	31 (53.5)	0.463
Missing knowledge about delirium	41 (45.1)	20 (60.6)	21 (36.2)	**0.025**
Patients who are difficult for assessment	38 (41.8)	9 (27.3)	29 (50.0)	**0.035**
Other problems are more challenging	20 (22.0)	3 (9.1)	17 (29.3)	**0.034**
Not enough motivated staff	19 (20.9)	10 (30.3)	9 (15.5)	0.095
No cost/resources for promoting at the department	19 (20.9)	7 (21.2)	12 (20.7)	0.953
Missing attitude, delirium is not important	18 (19.8)	6 (18.2)	12 (20.7)	0.773
Lack of nonpharmacological interventions	13 (14.3)	5 (15.2)	8 (13.8)	0.859
No appropriate scores for assessment of delirium	13 (14.3)	8 (24.2)	5 (8.6)	**0.041**
We have no barriers; delirium is regularly assessed	2 (2.2)	1 (3.0)	1 (1.7)	0.596
Leadership does not support	2 (2.2)	2 (6.1)	0 (0.0)	0.129
Lack of pharmacological interventions	2 (2.2)	0 (0.0)	2 (3.5)	0.533
Interprofessional conflicts	3 (3.3)	0 (0.0)	3 (5.2)	0.551
Missing data	0 (0)			

This table describes the reported barriers to delirium care from the reporting units.

- This study documented a delirium rate of 17%, so in 2021 there were 34 million hospitalisations meaning 5.8 million pt experienced delirium during their stay
- This is most likely an underestimate, previous studies have noted that upwards of 70% of delirium is missed in clinical care
- Best practice for delirium management, as supported by numerous meta analyses, is to avoid/minimise antipsychotics and benzodiazepines
- Despite this, over 80% of the units in this study disclosed the use of antipsychotics, and a third reported benzo use
- Benzos have been shown to inc the risk of delirium in ICU

Notes on AI in delirium

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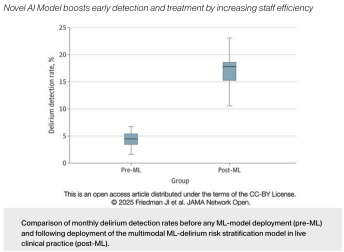
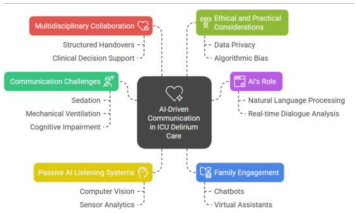
- Moreover, ML-analysis of electroencephalography (EEG) to predict post-operative delirium has now been tested [11]. Investigators obtained pre-operative resting EEG data from 85 patients undergoing elective surgery, used EEG patterns to generate a ML predictive model and tested its performance in an independent prospective cohort. They found a specificity of > 90% and a sensitivity of > 80% for a positive CAM assessment in the postoperative period.
- ^^However, the logistic and cost burden of EEG assessment makes use of this technology on a wide scale impractical at this time. In summary, even accounting for the above caveats, ML opens the door to early automated screening and intervention and may facilitate future trials of delirium treatment and rapid analyses of massive data sets.
- In a recent study, such combined ML-NLP analysis delivered an AUCROC > 0.917 and an area under the precision recall curve of 0.897, where delirium was diagnosed and agreed upon by two independent expert clinicians analysing all available documents, features, notes, and laboratory data [14]. Thus, the language of clinical context as captured in bedside notes can be merged with numerical data in large-scale epidemiological studies and learning models to inform diagnostic and therapeutic knowledge
- An AI model that can provide early warnings of clinical deterioration here
- the absence of diagnostic biological parameters and the fluctuating characteristics of delirium create a condition that lacks a diagnostic gold standard. In this setting, AI, large language models, ML and NLP all offer a real and novel opportunity to develop new operational standards for delirium prediction and diagnosis and open the door to better patient selection for interventional trials
- It quadrupled the rate of delirium detection and treatment



- This model identified pt at high risk, and then alerts a specialised to assess the patient
- The researchers used the AI model to analyze a combination of structured data and clinicians' notes from electronic health records. It used machine learning to identify chart data patterns associated with a high risk of delirium and applied natural language processing to identify patterns from the language of chart notes written by hospital staff. This approach captures staff observations of subtle mental status changes in patients who are delirious or at heightened risk.
- the implementation of an AI-based communication platform like the EyeControl-Med device has demonstrated promising results in facilitating communication and reducing delirium in ventilated ICU patients. Quantitative analysis showed a significant mean improvement of 19.3 points on the Loewenstein Communication Scale ($p < 0.0001$), while the mean number of errors on the CAM-ICU questionnaire decreased from 6.5 to 2.5 ($p = 0.0006$), indicating substantial reduction in delirium manifestations [47]. A large-scale randomized controlled trial (EMERGE) is currently underway to further evaluate this platform's efficacy in overcoming these critical communication gaps in delirium management
- transformer-based NLP models applied to clinician-family dialogues have been shown to identify subtle linguistic markers of evolving delirium 24–48 h before clinical manifestation, with studies reporting an AUC of 0.84–0.91 for early prediction [32], [33]. [27], [38]. This early detection capability enables timely non-pharmacological interventions, reducing delirium incidence by approximately 30 % compared to standard screening alone
- Passive AI listening systems enable early delirium detection by continuously monitoring ambient sounds, physiological parameters, and behavioral cues through advanced computer vision and sensor analytics [48]. These systems utilize computer vision algorithms to analyze visual data, detecting subtle alterations in speech patterns, facial expressions, and motor activity that may indicate delirium onset. This capability is particularly valuable in ICU settings where patients often cannot communicate directly. The operationalization of these systems involves sophisticated AI architectures that fuse multimodal perception stacks including facial key point estimation, gaze tracking, and inertial sensors with vision-language models to decode nonverbal patient cues into interpretable clinical intents [52], [53], [54]. Furthermore, streaming multimodal architectures integrate EHR data, physiological signals, and behavioral features within transformer-based models, enabling continuous, passive estimation of delirium risk trajectories [55], [56], [57].
- Based on a large-scale multi-center study of 104,303 patients from 195 hospitals, the DeLLiriuM model successfully predicted delirium risk. It demonstrated superior performance in external validation, achieving AUROCs of 0.77 (95 % CI 0.76–0.78) on the MIMIC-IV cohort and 0.84 (95 % CI 0.83–0.85) on the eICU cohort, outperforming all structured and text-based deep learning baselines. Key predictive features identified via SHAP analysis included urine specific gravity, ventilator settings (PEEP, tidal volume), and vital signs, aligning with established clinical risk factors. This study establishes DeLLiriuM as the first high-performing, broadly generalizable LLM for ICU delirium prediction
- Clinical validation demonstrates significant improvements in delirium management, with computer vision algorithms detecting facial expressivity changes and motor agitation achieving 89.25 % accuracy (95 % CI 88.75–89.73 %) for neurological disorder detection [58]. When integrated into clinical workflows, these passive monitoring systems reduced median time to delirium detection from 12.4 h to 2.1 h, enabling earlier intervention and reducing delirium duration by 1.8 days on average [59]. Complementing this technological approach, the Family Automated Voice Recording (FAVoR) intervention provides an effective non-pharmacological prevention strategy, significantly increasing delirium-free days to 34.8 % compared to 33.7 % with usual care ($\beta = 1.094$, $P < 0.001$) and demonstrating a strong dose–response relationship ($r = 0.62$, $P < 0.001$)[60].
- Evidence suggests that such AI tools can improve workflow efficiency and reduce clinician cognitive load. For instance, the AI-AntiDelirium system significantly improved nurses' adherence to delirium interventions (75 % vs. 58 %, $P < 0.01$), reduced extraneous cognitive load ($P < 0.01$), and enhanced germane cognitive load ($P < 0.01$)



https://www.sciencedirect.com/science/article/pii/S0964339725003854#b0125 This paper basically summarised a lot of the latter end of this, it's boss

- In this study, we introduce DeLLiriuM, a novel LLM-based delirium prediction model that utilizes EHR data from the first 24 hours of ICU admission to estimate a patient's risk of developing delirium for the remainder of their ICU stay. We developed and validated DeLLiriuM using ICU admissions from 104,303 patients across 195 hospitals in three large databases: the eICU Collaborative Research Database, the Medical Information Mart for Intensive Care (MIMIC)-IV, and the University of Florida's Integrated Data Repository. Our DeLLiriuM model achieved superior performance compared to all baseline models on the external validation set, measured by the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC) metric. DeLLiriuM attained AUROC 82.4 (95% confidence interval 81.8–83.0) and AUPRC 11.8 (95% confidence interval 11.3–12.4) across 77,543 patients spanning 194 hospitals.
- The hypoactive subtype of delirium is more common than hyperactive, but it's less likely to be discovered or reported, as these pt exhibit fewer behavioural problems and are often perceived as cooperative

Other links used:

https://link.springer.com/article/10.1007/s00134-024-07774-0

https://www.sciencedirect.com/science/article/pii/S2667296024000673?utm_source=chatgpt.com#bib43

https://pmc.ncbi.nlm.nih.gov/articles/PMC11887080/

https://pmc.ncbi.nlm.nih.gov/articles/PMC9839276/?utm_source=chatgpt.com

https://journals.healio.com/doi/10.3928/00989134-20080901-12 Read over this one

- Every hour, they get checked, they play a game that can help detect delirium and track progress

https://www.researchgate.net/publication/348680098_The_economic_cost_of_delirium_A_systematic_review_and_quality_as

Study on the economic costs of delirium:

- Increases nursing time per patient, higher subsequent rates of nursing home placement
- Delirium is shown to be avoidable in 30-40% of cases
- Delirium led to a 50% higher health-care cost, and a 37% higher cost when delirium was super-imposed on dementia
- The economic cost of delirium in the inpatient setting ranges between $806 and $24,509 per patient
- In intensive care unit it was between $1,529 and $14,462
- In community-based residents it was between $1,045 and $12,452
- inpatient delirium would cost somewhere between $6.6 billion and $82.4 billion in the United States alone (in 2019 US$)

- Study points out that one reason for delirium being poorly recognised in practice is the lack of a consistent application of clinically effective assessment tools, a range of tools are applied
- Another reason for the above is that there is no unifying definition of what defines an episode of delirium. TF inconsistent definition leads to inconsistent case detection
- Delirium has been linked with acceleration of dementia

https://onlinelibrary.wiley.com/doi/epdf/10.1002/gps.4690?getft_integrator=sciencedirect_contenthosting&src=getftr&utm_source=sciencedirect_contenthosting

Study on detection and management of hyper vs hypoactive delirium

Background:

- This is a retrospective study exploring daily hospital practice regarding detection and management of both hyper and hypoactive delirium of older patients during their hospitalisation
- Evaluated 65+ year olds diagnosed with delirium at Maastricht University Medical Center in 2014
- There are four subtypes of delirium: Hyperactive, hypoactive, mixed and without motor symptoms
- Hyperactive is characterised by agitation (can be motor), restlessness, aggressiveness
- Hypoactive is motor retardation, apathy, slowing of speech, sedated appearance
- Mixed is a combo of both
- No motor symp means pt only experiences cognitive symptoms of delirium
- Most commonly reported cause of delirium was infection (UTI, pneumonia etc) and surgery. Other causes were medication use and falls, in this study for 30% of pt there was no direct cause established
- 86% of pt received haloperidol to manage their delirium
- In this study, less than half of the patients could return back home after the hospital stay, and 15% died during hospitalisation
- A trend was observed of higher in hospital mortality among pt with a hypoactive delirium, also that they had longer length of stay
- In this study, hypoactive delirium was only identified in 25% of the pt whilst various studies find that hypoactive is the most prevalent, up to 56%
- Medical staff often overlook pt with hypoactive delirium as they are mostly passive and quiet, they're perceived as cooperative
- Studies have shown that many physicians think delirium is a harmless side effect of hospitalisation TF meaning that some physicians may have not put this on patient file pr discharge letter

▼ The role of AI-driven communication in delirium prevention, detection and care for critically ill ICU pt: a systematic review

https://www.sciencedirect.com/science/article/pii/S0964339725003854

Background:

- Became available online in January 2026, so it's a very recent paper
- It looked at studies exploring this area between Jan 2015 and May 2025

Intro

- Delirium is one of the most formidable and consequential complications affecting critically ill patients in ICU-prevalence ranges from 20-80% depending on pt population
- Delirium is associated with significantly prolonged mechanical ventilation duration, extended ICU and hospital stays

- Inpatient costs of delirium are $81 billion annually
- Mortality risk increases up to 3-fold in year following diagnosis
- Greater risk of long-term cognitive impairment
- Current approaches to manage delirium:
- Multicomponent interventions were most used in terms of non-pharm management, reduces delirium incidence by 27-54%
- ABC-DEF bundle is often used here
- Also involve environmental modifications like: promoting better sleep hygiene (noise reduction, light control), cognitive reorientation techniques (adding clocks or calendars to rooms), providing sensory aids like glasses or hearing aids.
- Physical training and reorientation protocols were single component interventions used
- Pharm management- antipsychotics like haloperidol or atypical antipsych, they're recommended to be used if severe and unresponsive to options above
- Confusion Assessment Method is screening often used in ICU
- Systemic and patient related barriers that cause suboptimal detection rates:
- High nursing workload, lack of routine screening adherence, inadequate training, sedation masks behavioural symptoms
- Delirium freq goes unrecognised, in up to 60-80% cases- communication barriers among pt, families and care teams cause this
- AI using EHR data- demographics, vitals, labs, have done well in predicting delirium
- Specific AI models have been validated for Post-op delirium in surgical cohorts
- Sedation and mechanical ventilation in ICU often impair patients' ability to verbalise symptoms, further complicating communication and clinical assessment, thus innovative solutions are needed to bridge these communication gaps

Results

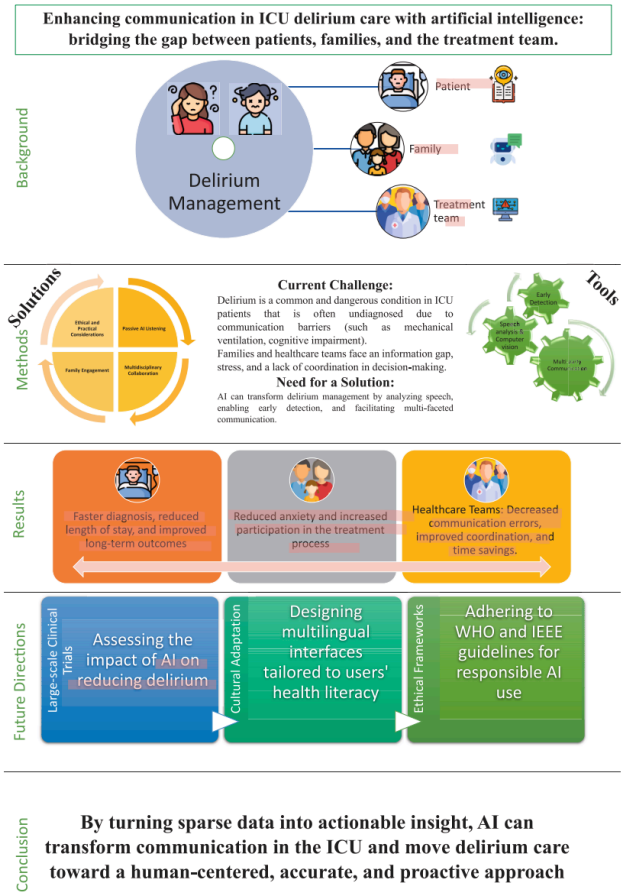


Fig. 1. AI-Enhanced Communication Framework in ICU Delirium Care. This visual abstract summarizes the potential transformative role of artificial intelligence (AI) in bridging critical communication gaps within ICU delirium care. It outlines the current challenges posed by communication barriers among patients, families, and clinicians; illustrates how AI-driven tools facilitate multidirectional communication for early detection, prevention, and management; and highlights the resulting improvements in patient outcomes, family engagement, and clinical teamwork. The framework emphasizes a future direction centered on responsible, ethical AI implementation guided by established international standards, ultimately promoting a more proactive, human-centered approach to delirium care.

- Six core thematic domains were derived from looking at all relevant studies:

Inherent communication barriers

- Sedated or ventilated pt are unable to communicate- whilst families face anxiety and exclusion from decisions
- This is compounded on other specific barriers
- AI-based communication platform- EyeControl-Med device has shown to facilitate communication and reduce delirium in ventilated ICU pt
- Lead to a reduction in mean number of errors on CAM-ICU questionnaire decreasing from 6.5 to 2.5
- Go into more detail about this on EMERGE trial

AI as multidirectional interface

- Bridges communication gaps amongst pt, families and MDTs
- Via natural language processing (NLP) AI transcribes and analyses clinical conversations in. real time and extracts critical insights and emotional cues to support delirium management.
- In this case it's transformer-based NLP models, they have a nuanced understanding of clinical dialogue and emotional tone
- Thanks to these models they have picked up on subtle linguistic markers of evolving delirium 24-48h before clinical manifestation, this reduced delirium incidence by 30%, AUC of 0.84-0.91

Passive AI listening for early detection

- Monitors ambient sounds, physiological parameters, behavioiural cues, all through advanced computer vision and sensory analytics
- These systems utilise computer vision algorithms to analyse visual data- detect changes in speech patterns, facial expressions and motor activity that could point towards deliirum onset
- Facial key point estimation, gaze tracking and inertial sensors with vision language models decode nonverbal pt cues
- Integreated with EHR data, it allows passive estimation of delirium risk
- Family Automated Voice Recording intervention was used here as a non-pharm prevention strategy- increased delirium free days to 34.8% from 33.7% ($P < 0.001$), it showed a strong dose-response relationship
- The way it worked is that the patients' family record voice messages reminding the patient, who they area, where they are, what day it is, reassurance, some memories etc
- These are played to the pt to help with reorientation, the more it was used the better the outcome
- The following were used

A) AI-driven family portals / dashboards

- Families get updates in plain language
- Education about delirium
- Prompts for what they can do (reorientation, mobilising, bringing hearing aids/glasses, sleep cues)
- A way to send observations back to staff (bidirectional)

B) Multilingual chatbots using RAG

- RAG means the chatbot can answer questions using the hospital's approved content (policies, patient-specific info, delirium guidance) rather than "making it up".
- Multilingual + different literacy levels = families understand and engage more.

C) "Voice-enabled family agents"

This does **not** have to mean voice cloning or the agent "speaking as the family member."
It usually means:

- The family can interact with the system **by voice** (like calling an assistant), or
- The system can capture family updates/concerns via voice, or
- It can read summaries aloud.

D) Ambient intelligence: diarization + sentiment analysis

- **Diarization** = automatically identifying who is speaking (clinician vs family) in a conversation.
- **Sentiment analysis** = detecting emotional tone/distress.
- Used to generate summaries and flag unmet psychosocial needs (e.g., family confusion, anxiety, disagreements, unanswered questions).

- ^^ This is actually related to the next theme of AI-enhanced family engagement
- AI powered family portals enabled a 34% increase in family delivered non pharm interventions and a 29% reduction in delirium incidence + a 1.5 day shorter delirium duration among patients

AI structured handovers for teamwork

- Communication breakdowns are common in high pressure ICU
- AI comms tools streamline handovers, priortise clinical info and coordinate care activities
- AI-AntiDelirium system significantly improved nurses adherence to delirium interventions to 75% from 58%
- An AI reduced delirium incidence from 47.7% to 22.3% in surgical ICU pt
- ABCDEF bundle reduced delirium prevalence from 33.48% to 28.74% and a 7.58% reduction in midazolam

Discussion

Ethical-regulatory-practical challenges

- Knowledge-freezing effect- trained model's static knowledge base not keeping up with dynamic nature of medical evidence TF can't adapt?
- Lenghty clinical validation and approval processes
- Privacy and ethical concerns, pt worry about data security and potential misuse of sensitive health info

▼ Validation of the Prediction of Delirium for Intensive Care model to predict subsyndromal delirium

https://onlinelibrary.wiley.com/doi/10.1002/ams2.378

- Risk of delirium is elevated in elderly and physically weak individuals
- Subsyndromal delirium (SSD) is where the condition doesn't satisfy the diagnostic delirium criteria of DSM-5
- It's an intermediate stage between delirium and normal mental status
- Intensive care delirium screening checklist is used, if between 1-3 it's SSD
- A cause of inc mortality and prolonged stay - SSD
- The average medical ICU pt has 11 or more risk factors for developing delirium
- Prediction of Delirium for Intensive Care (PRE-DELIRIC) is a model for predicting the incidence of delirium in adult ICU inpatients
- Made up of 10 risk factors
- AUROC = 0.87
- Study was between May 2015 and Feb 2017 in Tokyo
- The pathophys of delirium is thought to be associated with sleep disturbances in critically ill pt
- Sleep hygiene programs in ICUs improve sleep quality and reduce incidence of delirium
- Attention disturbance is a core delirium symptom, it's caused by dysfunction in the ascending reticular activating system, which regulates arousal

▼ Postoperative delirium prediction using ML models and preop EHR data

https://link.springer.com/article/10.1186/s12871-021-01543-y?utm_source=getftr&utm_medium=getftr&utm_campaign=getftr_pilot&getft_integrator=scopus

- 115 preop risk features including demographics, comorbidities, nursing assessments, surgery type and other preop EHR data were used to predict post op delirium (POD)

  Supplementary Table S1: List of variables included in the machine learning models

Category	***Variable***
Patient characteristics and demographics	Age Gender Race Ethnicity Primary language Interpreter needed Marital status Insurance type Need for transportation Food insecurity Financial resource strain Emergency contact

	available Admission status Admission source Billing source Preoperative location
Medical history	Allergies Number of allergies Number of severe allergies Past medical history (ICD 10 codes) Number of prior anesthetic encounters in the past 3 years Smoking status Alcohol use Substance use Prior admission 30 days Prior admission 90 days ASA Class
Surgical characteristics	Day of surgery Month of year Procedure name ERAS patient (yes/no) Predicted case length Case location Time of induction Delayed incision (yes/no) Emergent case Case classification Admitting service Surgical service Planned anesthetic type
Preoperative nursing assessments	Learning preference Barriers to learning (visual, hearing, cognitive, etc) Type of pain assessment (ex: numeric, faces, CPOT) Pain or discomfort Pain level (0-10) Acceptable level of pain Type of pain Location of pain Area of pain Effect of pain on daily activities Multiple areas of pain (yes/no) Pain relieving factors Pain treatment strategies Thermoregulation interventions Respiratory interventions Ability to spell ‘WORLD’ backwards Illness severity score Orientation to place Access to transportation ADL: feeding ADL: grooming ADL: bathing

	ADL: dressing ADL: bed transferring Weakness in arms Weakness in legs Bowel and bladder habits Vision issues Hearing issues (R ear) Hearing issues (L ear) Mobility issues Elimination issues History of falls Schmid Fall score Braden scale Fall risk stratification (0-5) Problem documented: Neuro Problem documented: Cardiac Problem documented: Skin Problem documented: Musculoskeletal Problem documented: Genitourinary Problem documented: Anus and rectum Problem documented: HEENT Problem documented: Psychosocial Violence abuse assessment Social worker consultation needed Suicide risk Unplanned weight loss Difficulty chewing Difficulty swallowing Tube feeds Total parenteral nutrition Presence of pressure ulcer Non healing wound Dietician consultation needed Oxygen device Braces / devices / sensory aids Sleep habit details Glasgow Coma Score
Vital signs	Temperature Pulse rate Source of pulse Respirations Oxygen saturation Blood pressure Source of blood pressure Weight (kg) BMI Height (cm) Body surface area
Labs	Glucose

***Abbreviations**: ICD 10, International Classification of Diseases, 10th revision; ASA, American Society of Anesthesiologists; ERAS, enhanced recovery after surgery; CPOT, critical care pain observation tool; ADL, activities of daily living; HEENT, head eyes ears nose throat; kg, kilogram; BMI, body mass index; cm, centimeter*	

- The ML models outperformed traditional approaches in detecting POD
- 2 ML models were discussed- XGBoost and Neural Net
- **XGBoost AUC-ROC ≈ 0.851**
- **Neural network AUC-ROC ≈ 0.841**
- **AWOL-S AUC-ROC ≈ 0.762**
- AWOL-S is an EHR-based risk stratification tool in which regression coeffs are applied to each of five terms- Age, ability to spell WORLD backward, Orientation to place, iLlness severity and prcedure specific Surgical risk
- POD is associated with numerous adverse events including prolonged length of stay, higher re-admission rates, dependency in activities of daily living and cognitive decline
- Study looked at pt 18+ having surgery Between 2016-2019
- Another benefit of the ML models vs traditional ones is that they predict delirium in a broad surgical population over a wide age range vs classic ones that focus on one particular surgical population

▼ (very similar to above)- AI-delirium guard: predictive modeling of POD in elderly surgical pt

- https://pmc.ncbi.nlm.nih.gov/articles/PMC12140226/pdf/pone.0322032.pdf
- This model just reinforces the previous article- talks about how XGBoost model was superior- AUC of 0.85

▼ EyeControl Communication Platform for Delirium Management in ICUs

- https://www.sciencedirect.com/science/article/pii/S2949788426000213
- Results aren't out yet for this paper, I'm assuming they're still collecting the data
- Delirium is an acute, fluctuating disturbance of consciousness and cognition
- Prevalence ranges from 60-80%
- ICU delirium in a separate study was the strongest independent determinant of hospital LOS
- Each additional day in delirium was estimated to inc risk of prolonged hospitalisation by 20% and mortality by 10%
- Meta analyses consistently show that antipsychotics offer no benefit yet remain widely used
- Compliance with bundles has been shown to reduce burden of delirium in a dose dependent manner
- Consistent implementation of the components of the bundle is the issue
- EyeControl-Pro is a communication platform for ICU delirium
- Delivers one-way sensory stimulation through periodic orientation messages, music and family messages to noncommunicative critically ill pt
- A single center pilot study on this showed a 19.3 inc in the Loewenstein communication scale reduced mean number of errors on CAM-ICU from 6.5 to 2.5

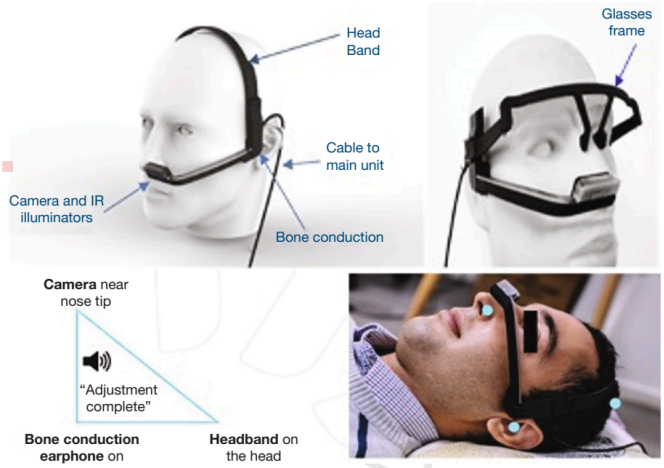


Figure 2 – *Images showing the headset. The head unit can be mounted to the patient's head using 1 of 2 methods: a glasses frame or a headband. The glasses frame is the preferred use for patients in bed, but when needed can be replaced by a headband (because of patient injury, other necessary devices mounted, or if a patient must lie on the side of the head).*

- Has a head unit with an infrared camera to track eye movements and bone conduction earpiece for audio feedback
- Has a dashboard connected that allows clinicians to monitor pt status remotely including eye state (open or closed), communication attempts
- Family members can upload personalised messages securely that are played through the earpiece at periodic intervals
- Plays orientation messages and music from a wide selection of genres
- It also performs automated CAM-ICU assessments via a voice interface, pt responds with eye blinks

▼ Epidemiology and assessments of delirium in nursing homes and rehab facilities

- https://pmc.ncbi.nlm.nih.gov/articles/PMC12743668/pdf/41999_2025_Article_1207.pdf
- Explores delirium prevalence and assesment methods in LTC facilities across multiple countries in nursing homes vs rehab facilities
- 92,005 older adults in LTC in Canada (a separate study) showed delirium significantly affected both patients' prognosis and cognitive trajectory at a one year follow-up
- Delirium in LTC residents has been associated with inc risk of falls, more rapid functional decline, frequent hospital admissions
- These pt are frail and cognitively impaired, both are well-established predisposing factors for delirium onset
- Rehab facilities investigated it at least once a day whilst the majority of nursing homes only checked it in a case of sudden consciousness changes
- In rehab facilities it was predominantly nurses doing the checks, in nursing homes it was physicians
- Rehab facilities reported more freq communication about delirium in handovers
- Rehab facilities showed a higher prevalence of delirium, but this could be because delirium assessments are carried out more frequently
- Previous studies show family members play a significant role in early recognition of delirium as thye can detect subtle cognitive changes

▼ A thesis on analysis of delirium:

- Inflammatory markers were associated with delirium status- ferritin and neutrophil counts

- Neutrophil:lymphocyte ratio (NLR) was higher in those with 4AT ≥ 4 than those with 4AT 0
- Frames delirium through a neuro-inflammatory hypothesis- systemic inflammation (from acute illness/infection/surgery) can drive brain dysfunction via mechanisms such as BBB disruption and neuroinflammation
- Higher WCC/neutrophils/ferritin/CRP and lower albumin/lymph/platelets in higher 4AT groups
- Systemic inflammatory response syndrome severity- a severe whole body inflammatory state caused by infection, trauma, burns, was a statistically significant inflammatory predictor for delirium
- Low lymph + high neutrophils → high NLR → associated with delirium being more likely
- Within 4AT ≥ 4 group, markers like NLR > 5 and abnormal WCC were strong predictors of 30 day inpatient mortality. This helps us show that inflammatory physiology can help identify who is at highest risk

▼ NLP techniques to detect delirium via clinical notes

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